

Design and Development of an Agentic System for Context-Aware Daily Mobility Services

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Abstract— Harnessing advances in agentic artificial intelligence, real-time mobility services, and vehicle automation creates new opportunities to coordinate household travel as a continuous daily mobility service rather than a series of isolated trips. This demonstration presents an agentic mobility assistant that combines bounded large language model (LLM) reasoning with deterministic calendar processing, traffic-aware routing, construction verification, electric-vehicle charging coordination, personalized output generation, and persistent route memory. The system constructs a continuous household vehicle mission from calendar and vehicle context, verifies operational details using external mobility services, and stores the resulting state for later follow-up interaction. The demonstration shows two representative household scenarios. Run 1 is a feasible six-trip day with an 80% battery, sequential caregiver drop-offs, a caregiver pickup, two protected school pickups, and no unnecessary charging or schedule intervention. Run 2 reproduces the constrained Montréal evaluation case, where low battery, a conflicting medical appointment, and construction require charging, schedule adaptation, and rerouting. The result is a persistent mobility service that can plan, verify, remember, and explain household transportation decisions.

Introduction

Modern households face growing mobility-coordination challenges—more complex daily schedules, increasingly time-sensitive caregiving responsibilities, and greater pressure to efficiently share limited transportation resources. A typical weekday may involve overlapping work commitments, medical appointments, school pickups, caregiving responsibilities, and vehicle-related tasks. When the household relies on one shared vehicle, a useful assistant must preserve high-priority responsibilities while maintaining vehicle continuity, passenger continuity, feasible departure times, and sufficient energy for the complete day.

Prior personalized travel-planning systems incorporate user schedules and preferences, while situation-aware routing shows that route choice depends on traveler and environmental context (Chiang and Huang 2015; Campigotto et al. 2017). Recent agentic systems extend travel assistance by coordinating LLM reasoning with external traffic, charging, and planning tools (Khamis 2025; Chen et al. 2026). However, these approaches primarily focus on individual users or itineraries rather than a persistent household mission that coordinates multiple calendars, caregiver assignments, protected dependents, charging, construction, user-specific visibility, and vehicle-facing operations through one shared vehicle. The present demonstration focuses on this integrated service behavior. The contribution of this demonstration is the integration of persistent household mission planning, deterministic operational verification, separate user- and vehicle-facing mission views, and route-memory interaction within one shared-mobility service.

System Overview

The prototype separates contextual reasoning from externally verifiable mobility processing. Bounded LLM agents classify requests, reason about household responsibilities and scheduling conflicts, interpret calendar modifications, and synthesize personalized responses. Deterministic workflow components retrieve and normalize calendars, validate destinations, geocode locations, generate traffic-aware route legs, verify construction impacts against route geometry, evaluate charging opportunities, filter information for the active user, and store the complete mission in persistent route memory. This separation prevents the LLM from independently inventing coordinates, travel times, route geometry, calendar records, or vehicle-state facts.

Execution may begin from either a user request or a scheduled daily pre-computation trigger. A new planning execution first determines the household mission, then verifies its operational route, timing, charging, and environmental details, and finally produces two views of the same state: a personalized user-facing route and a complete vehicle-facing operational plan. The verified mission is indexed by user and local planning date. Subsequent follow-up questions retrieve that state without repeating calendar retrieval, route generation, construction analysis, or vehicle-resource processing unless an update is required.

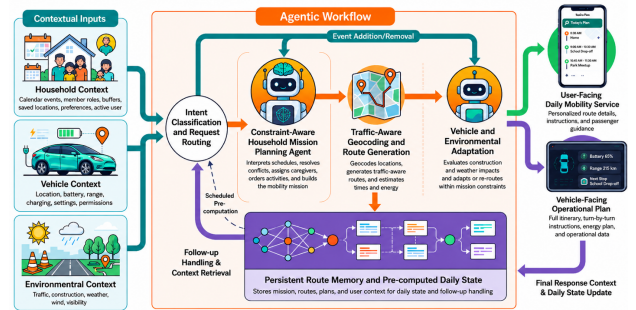


Figure 1: Agentic mobility service architecture. Household, vehicle, and environmental context are coordinated through bounded LLM reasoning, deterministic mobility processing, and persistent route memory to produce user- and vehicle-facing outputs.

Demonstration

The live demonstration uses a conversational interface connected to the workflow through webhooks. Household schedule information is provided by Google Calendar; TomTom services support geocoding and traffic-aware route generation; Montréal construction records are used for roadwork verification; Circuit électrique information supports charging decisions; and weather data are retained as contextual information. Before Run 1, the interface exposes the household locations, vehicle battery thresholds and return-home behavior, arrival buffers, construction-avoidance limits, and weather-aware planning settings that shape the generated mission. The demonstration emphasizes observable system behavior rather than the internal workflow canvas.

Source code, setup instructions, and implementation materials are available at:

<https://github.com/olibreadstick/agentic-mobility-assistant>

Representative Run 1: feasible shared-vehicle day.

The first run demonstrates a feasible household day with two caregivers, two dependents, one shared electric vehicle, and Parent1 as the active user. Parent2 has a 9:00 AM work meeting at 1 Place Ville Marie, Parent1 has a 10:00 AM research meeting at the Macdonald Engineering Building, Child1 has a protected 3:30 PM school pickup, and Child2 has a protected 4:15 PM school pickup. The vehicle begins at 80% battery, above the preferred 65% threshold, so no charging intervention is required. A direct planning request reconstructs the mission from the current household, vehicle, and routing context.

The user-facing plan contains six driving trips: Home → Parent2 workplace → Parent1 workplace → Parent2 workplace for caregiver pickup → Child1 school → Child2 school → Home. The first two legs preserve shared-ride continuity by dropping Parent2 first while Parent1 remains in the vehicle. Parent1's pickup after the research meeting occurs at the same location and is represented as an on-site transition rather than an unnecessary zero-distance driving leg. Parent1 then completes both protected school pickups before the final return home. This run shows that the workflow can coordinate caregiver drop-offs and pickups, protected dependent responsibilities, passenger continuity, traffic-aware departure times, and battery-aware decisions without introducing unnecessary charging or rescheduling.

Representative Run 2: constrained Montréal evaluation case.

The second run reproduces the constrained Montréal scenario used in the accompanying evaluation. The shared vehicle begins at 35% battery, below the preferred 65% threshold. Parent1 is responsible for two protected school pickups while a 4:00 PM medical appointment conflicts with the pickup sequence. The workflow preserves the dependent responsibilities, inserts an autonomous charging action during an available idle period, and recommends rescheduling the conflicting medical event rather than moving the protected pickups. In the complete vehicle-facing plan, the system produces an eight-leg mission covering approximately 17.9 km and 1 h 19 min of estimated driving. Construction on Rue Mansfield is checked against the preliminary route geometry; after overlap is confirmed, an alternative through McGill College Avenue is retained because the detour remains within the configured rerouting limits. In contrast with Run 1, this scenario demonstrates active intervention under energy, scheduling, and environmental constraints.

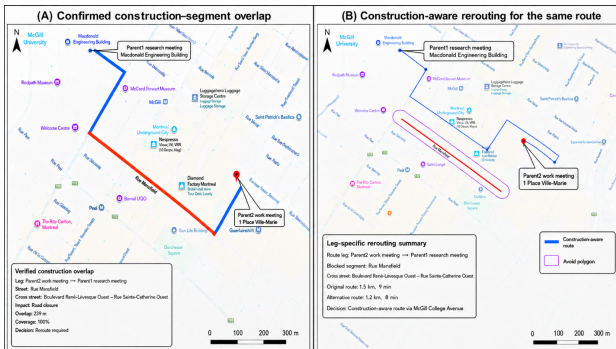


Figure 2: Construction-aware adaptation in Run 2. The preliminary route overlaps the reported Rue Mansfield work zone; the retained alternative avoids the affected segment through McGill College Avenue.

Persistent interaction.

After either mission has been stored, the interface can answer route-memory questions such as asking for the current plan, the departure time for a protected pickup, or the vehicle battery level. These requests are classified as follow-up

interactions and retrieve the stored mission instead of executing the full planning pipeline. This behavior demonstrates that persistent memory is not only an implementation cache: it is part of the service model, allowing the same verified daily state to support both a structured route dashboard and concise conversational answers.

Interactive capabilities and scope.

The interface supports five service intents: new route planning, route follow-up, calendar-event addition, calendar-event removal, and casual interaction. The demonstration concentrates on the two mobility paths most relevant to persistent planning. A new route request activates the complete mission workflow, whereas a follow-up request uses the stored route when it is valid for the same active user and local date. The user-facing dashboard presents the route date, contextual weather, schedule conflicts and recommendations, ordered trip cards, caregiver responsibilities, battery state, and charging information. Short follow-up questions can then retrieve specific facts such as departure times or battery level without redisplaying or recomputing the full mission.

The complete vehicle-facing plan is preserved separately from the personalized interface so that user-specific filtering does not remove movements required for vehicle operation. This output includes passenger-carrying trips, passenger-free movements, charging actions, caregiver pickups, dependent pickups, and return-home operations. In the present prototype this vehicle-facing plan is simulated workflow output rather than direct control of a deployed autonomous vehicle; live telemetry and physical vehicle execution remain future integration targets.

Conclusion

The demonstration shows how a conversational trip-planning assistant can be extended into a persistent daily mobility service for a shared household vehicle. Run 1 demonstrates that the system can coordinate a feasible six-trip day without unnecessary intervention, while Run 2 shows how the same architecture responds when low battery, a protected-pickup conflict, and construction require active planning decisions. Across both scenarios, the system coordinates interdependent household obligations, grounds operational quantities in deterministic mobility services, maintains separate user- and vehicle-facing views of a shared mission, and supports concise follow-up interaction through persistent route memory.

In controlled architectural comparisons using GPT-5.2, the complete system achieved 94.44% mean route accuracy, compared with 75.00% for an intent-guided LLM-only baseline and 72.22% for a calendar-tool agent. The largest differences appeared in verified alternative recommendations, active-user filtering, and schedule feasibility. Although the reduced systems were faster, they omitted traffic-aware routing, construction verification, charging coordination, user projection, and persistent route memory. These results indicate that the surrounding hybrid workflow contributes operational capabilities beyond high-level calendar reasoning alone. Its significance lies in connecting LLM-based household reasoning with verifiable mobility operations and persistent state, supporting agentic AI as a context-aware household mobility service.

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